

Application shell

operational program boundary

config file, generated command line, logging, daemonization, quiet mode

diagnostic evidence: effective configuration, output mode, startup state



Instance scope

configured server/client role visible at runtime

enabled/disabled state, listen/connect activation, retry scheduling

diagnostic evidence: role lifecycle, endpoint selection, activation result



Connection scope

one concrete peer relationship

connection name, local/remote addresses, queued/sent/read/processed data

diagnostic evidence: peer episode, counters, duration, disconnect summary



Context scope

protocol behavior attached to one connection

protocol state, accepted/rejected frames, application-level events

diagnostic evidence: protocol meaning, state transitions, close decisions

Logging forms

LOG(level)
ordinary lifecycle
and severity

PLOG(level)
lifecycle plus captured
system-error context

VLOG(n)
optional diagnostic
depth